Replicating: HPCC: High Precision Congestion Control

Team Members:
Andrea Bellani (andrea1.bellani@mail.polimi.it);

Andrea Migliorini (andrea1.migliorini@mail.polimi.it).

Source Paper:

Y. Li, R. Miao, H. H. Liu, Y. Zhuang, F. Feng, L. Tang, Z. Cao, M. Zhang, F. Kelly, M. Alizadeh, M. Yu: HPCC: High Precision Congestion Control. In SIGCOMM '19, August 19–23, 2019, Beijing, China, © 2019 Association for Computing Machinery.

Project: link to our repository

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1. Introduction

Introduce the paper by summarizing:

- The problem the paper addresses and its importance
- The key ideas behind its solution and its approach
- The main contributions

HPCC: High Precison Congestion Control presents a novel approach to Congestion Control for Datacenter networks.

1.1. The problem HPCC wants to address

HPCC aims to improve congestion control in datacenters high-speed networks. By focusing on Congestion Control in RDMA networks, it tries both to address the well-known problems that motivate the usage of congestion control mechanisms in RDMA networks and to overcome limitations of state-of-art Congestion Control mechanism for RDMA networks (in particular, DCQCN and TIMELY): * well-known problems that require congestion control in RDMA networks: - throughput maximization; - latency minimization; - high burst-tolerance; * main limitations of state-of-art Congestion Control mechanisms for RDMA networks: - easy-to-deploy; - slow convergence due to leveraging of heuristic approaches; - difficult parameters tuning.

1.2. Key ideas behind HPCC

The essential feature that HPCC leverages to out-perform its competitors is INT (In-Network Telemetry):

In fact, using INT enables HPCC to obtain precise flows information which other state-of-art CC mechanisms usually aquire with a much slower convergence.

HPCC is at his core a window-based congestion Control algorithm that keeps monitoring the number of inflight bytes.

1.3 Brief explanation of HPCC

HPCC main contributions

2. Selected Result

2.1. Fairness and queue size with W_{AI}

Considering an incast of 16 flows, Figure 14a shows the average throughput for different W_AI values, Figure 14b shows length's percentiles of the incast Queue

2.2. Different ways of reacting to ACK.

Considering an incast of 16 flows, Figure 13a and Figure 13b shows respectively the aggregated throughput and Queue Length running HPCC with different reacting policies

3. Environment Setup

Note: This section should contain enough information to allow someone else to reproduce *your* report. Share hardware and/or software setup relevant to your experiment. For example:

Hardware Environment: CPU, Memory, Storage, Network, Cloud / local / cluster, Any relevant micro-architectural details

Software Environment OS version, Kernel version, Compiler version, Libraries, Dependencies, Paper artifact used (Yes/No; version/commit hash)

Configuration Parameters:

- Workload configuration
- Dataset
- Runtime parameters and flags

Deviations from the Original Setup:

Clearly describe any difference between papers and your experiment environment.

- Hardware differences
- Software version differences
- Dataset substitutions
- Unavailable components

Explain why these deviations were necessary.

If something was **missing in the original paper**, state it. For example:

> The paper does not specify X. We assumed Y (or explored range a to b).

4. Experiment Result

> Explain how your experiment was conducted and then what results you acquired. Afterwards, compare your results with those of the paper and state your takeaways.

Step-by-step description:

1. Execution procedure
2. Measurement method
3. Number of runs
4. Statistical treatment (mean, median, CI, etc.)

Also Describe:

- How did you ensure correctness (did you check also other metrics to make sure the experiment is running correctly?)
- Did you do any debugging? Discuss issues you faced and how you overcame them (if applicable consider allocating a subsection for this item)

Share your result and compare them with the paper's. Then discuss your takeaways.

For comparison include:

- Graph(s) or table(s)
- Matching axes and units with the source paper
- Error bars if applicable
- You may want to report difference with the original results (e.g., absolute number or percentage).

For example:

```
<img
  alt="The figure shows that method A improves throughput compared to method B"
  src="figures/one_bar.png"
  style="width:100%"
  />
<p>Figure 2: The figure shows that method A improves throughput compared to method B</p>
```

```
<img
  alt="Our reproduction of Figure 1 shows results with the similar trend as claimed by the p
  src="figures/two_bar.png"
  style="width:100%"
```

/>
<p>Figure 3: Our reproduction of Figure 1 shows results with the similar trend as claimed by

> **Reminder:** the goal is not achieve the exact results of the paper, but to do a rigorous experiment with similar assumptions from the source paper and gain insight. The insight can be correctness of work, failure to reproduce same results, or even infeasibility of doing such experiment for interesting reasons.

5. Further Exploration

In this project you are required to also explore a research question of your own. Either:

1. Take the same test with different input workload or a variation of a test that is not present in the paper and comment the results you obtain
2. Implement a new feature on top of the system you evaluated and show a figure showing the performance

Discuss which approach you take, and what you explored. Explain what was your motivation and importance of your question.

5.1. Methodology and Result

Report the experiment you designed for answering the question and share the result you got.

Include:

- Graph(s) or table(s)
- How the experiment was conducted (share the details)
- What did you discover?

6. Reproducibility Assessment of the Paper

Evaluate the paper itself:

- Was the methodology clearly described?
- Was the artifact usable?
- How difficult was reproduction?

7. Conclusion

Conclude the report by mentioning the takeaways of experiments you did

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Appendix

You are asked to write this report using Markdown. You can find a cheat sheet of Markdown syntax at this link.

For generating a PDF file from your report you can use a tool of your choice. *md2pdf* is one such tool. See this link for more information about it. You can also use an online editor such as this.